

Bingo Game

Implementation of Bingo Game using JavaScript — Lab Report

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Subject	Web Application Development Lab
Exercise	3 — Implementation of Bingo Game using JavaScript
Year / Semester	III Year / V Semester
Batch	2024–2029

Objective

To design and implement a two-player Bingo game using HTML5, CSS3, and vanilla JavaScript, demonstrating DOM manipulation, event handling, and game-logic implementation (grid generation, turn management, and win detection).

Features

- Two independent 5×5 Bingo cards (Player 1 and Player 2), each filled with a randomly shuffled arrangement of numbers 1–25 (no duplicates).
- “Start” button generates fresh, randomized cards and begins the game.
- “Reset” button clears the game completely and returns it to its initial state.
- Turn-based play: players alternate entering numbers, with the current turn clearly displayed (“Player 1’s Turn” / “Player 2’s Turn”).
- A number entered by either player is marked on both cards simultaneously if present.
- Automatic win detection after every valid mark — checks all rows, all columns, and both diagonals.
- The winning line is visually highlighted (gold glow) on the winner’s card.
- Clear winner announcement, e.g. “Player 1 Wins! BINGO!”
- Game locks after a win — no further marking is allowed until Reset is pressed.
- Full input validation (see Validation Handled below).
- Pressing Enter in the number field also triggers Mark; input auto-focuses after each turn for fast play.
- Responsive layout — cards stack vertically on smaller screens.

Technologies Used

- HTML5
- CSS3 (Flexbox, CSS Grid, media queries, animations)

- Vanilla JavaScript (ES6) — no frameworks or libraries

Project Structure

```
bingo-game/
|-- index.html    # Page structure and layout
|-- style.css     # Green-themed, responsive styling
`-- script.js     # Complete game logic
```

How to Run

- 1 Extract the project files from the provided zip archive.
- 2 Open index.html directly in any modern web browser (Chrome, Firefox, Edge).
- 3 No build step, server, or internet connection is required — everything runs client-side.

How to Play

- 1 Click Start — two randomized 5×5 cards are generated, one for each player.
- 2 Player 1 enters a number between 1 and 25 in the input field and clicks Mark (or presses Enter).
- 3 If that number exists on Player 1's and/or Player 2's card, it is marked on both.
- 4 The turn automatically switches to Player 2, who repeats the process.
- 5 Play continues, alternating turns, until someone completes a row, column, or diagonal.
- 6 Click Reset at any time to clear the board and start a brand-new game.

Winning Condition

A player wins as soon as all five cells in any of the following are marked on their own card:

- Any complete row, or
- Any complete column, or
- Either diagonal (top-left → bottom-right, or top-right → bottom-left)

The winning line is highlighted, a “Player X Wins! BINGO!” message is shown, and further marking is disabled until Reset is clicked.

Validation Handled

Scenario	Message Shown
Mark clicked before Start	“Please start the game first.”
Input left empty	“Please enter a number.”
Non-numeric input	“Please enter a valid number.” (also blocked natively by the number input)
Number outside 1–25	“Enter a number between 1 and 25.”
Number already marked	“Number already marked.”
Marking attempted after game over	Blocked (Mark button and input disabled)

Invalid input never changes whose turn it is — only a successful mark advances the turn.

Testing

The application was built and then actually tested in a real browser (via automated Playwright scripts driving the generated index.html), covering:

- Initial UI state (Start enabled, Mark/input disabled, placeholder cards shown)
- Card generation: 25 cells per card, each containing all numbers 1–25 exactly once, for both players
- All invalid-input cases and their exact messages
- Confirmation that invalid input does not change the turn
- Marking a valid number updates both cards and switches the turn
- Duplicate-number handling and turn preservation
- Win detection for a row, a column, the main diagonal, and the anti-diagonal (tested independently)
- Winning-line highlighting and winner message text
- Game lock (Mark/input disabled) after a win
- Reset restores the initial state fully, and a new game can be started afterward
- Enter-key shortcut for Mark
- Responsive layout at a mobile viewport width (cards stack vertically)

All test cases passed with no console or JavaScript errors.

Bonus Features Implemented

- Smooth mark and winning-cell animations (scale/pulse effects).
- Auto-focus on the input field after every turn for faster gameplay.
- Enter-key support to trigger Mark without needing the mouse.
- Responsive design that adapts cleanly to mobile screen widths.

(An AI opponent and sound effects were not implemented, to keep the core two-player assignment requirement simple and fully working, as prioritized in the assignment brief.)